

Create and Connect to a Linux Virtual Machine (EC2 Instance)

[Edition 20]

[Last Update 231210]

Scenario Overview:

Embark on a practical journey to create and connect to a Linux EC2 instance on Amazon Web Services (AWS). This hands-on lab guides you through the process of setting up a virtual Linux server in the cloud, configuring security measures, and establishing a secure connection for remote access.

This lab offers a hands-on exploration into the creation and connection to a Linux EC2 instance, imparting essential skills for cloud-based computing. It provides a foundational understanding of AWS infrastructure, Linux virtualization, and the principles of secure remote access using SSH key pairs.

For any issues/help contact in Training WhatsApp Group or Support Ticketing System

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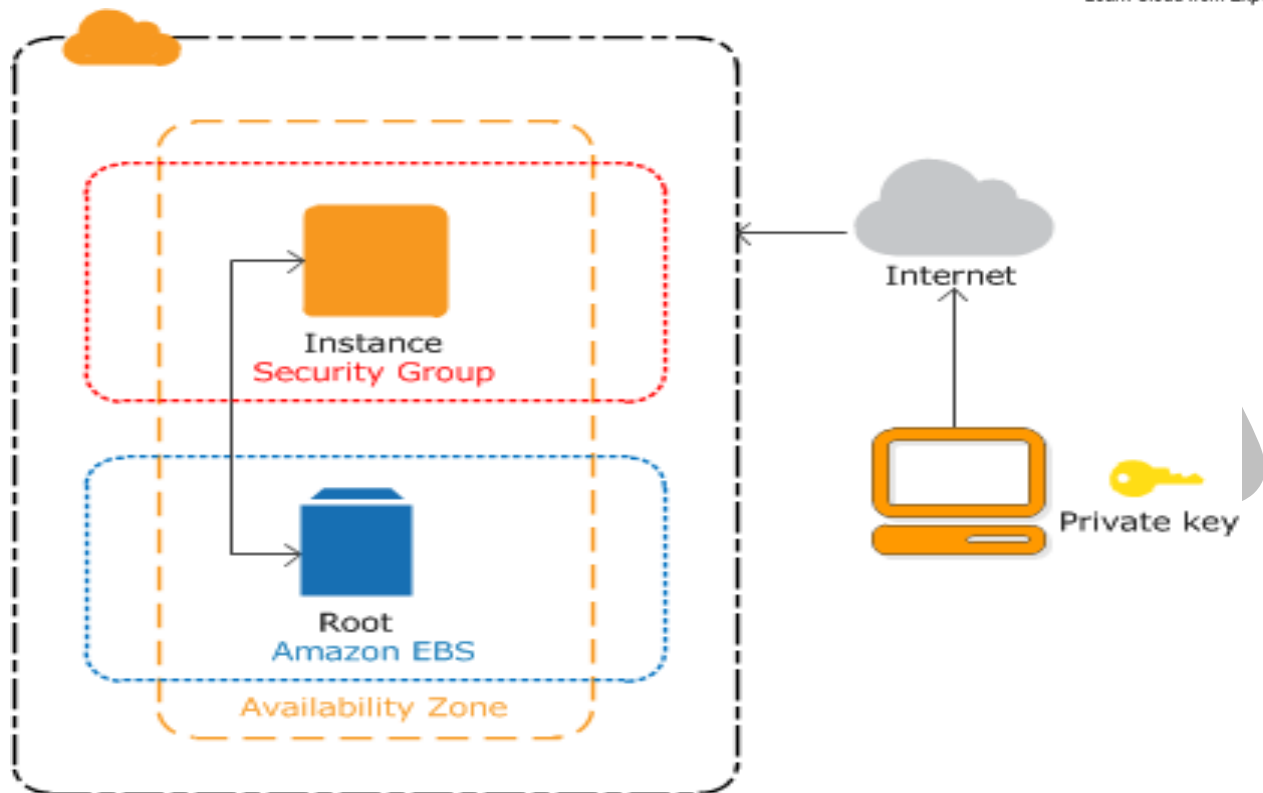
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1 INTRODUCTION

Amazon EC2 presents a true virtual computing environment, allowing you to use web service interfaces to launch instances with a variety of operating systems, load them with your custom application environment, manage your network's access permissions, and run your image using as many or few systems as you desire.

To use Amazon EC2, you simply:

- Select a pre-configured, templated Amazon Machine Image (AMI) to get up and running immediately, or create an AMI containing your applications, libraries, data, and associated configuration settings.
- Configure security and network access on your Amazon EC2 instance.
- Choose which instance type(s) you want, then start, terminate, and monitor as many instances of your AMI as needed, using the web service APIs or the variety of management tools provided.
- Determine whether you want to run in multiple locations, utilize static IP endpoints, or attach persistent block storage to your instances.
- Pay only for the resources that you actually consume, like instance-hours or data transfer.



This activity guide cover steps for:

1. Launching a Linux Instance using EC2
 - a. Creating an EC2 Instance
 - b. Accessing EC2 Instance Using Browser SSH Connection
 - c. Accessing EC2 Instance Using Putty
2. Deleting/Cleanup
 - a. Stopping the Linux Instance
 - b. Terminating the Linux Instance

2 OVERVIEW & KEY CONCEPTS

In this section we'll cover key concepts used in this guide, including an overview of what's covered in this guide.

2.1 What is EC2?

Amazon Elastic Compute Cloud (Amazon EC2) delivers scalable computing power on-demand in the Amazon Web Services (AWS) Cloud. Using Amazon EC2 lowers hardware expenses, allowing you to create and deploy apps quicker. Amazon EC2 allows you to deploy as many or as few virtual servers as you need, set security and networking, and manage storage. You can add capacity (scale up) to address compute-intensive operations such as monthly or annual procedures or surges in website traffic. When consumption falls, you can lower capacity.

The EC2 instance is located in a Region Availability Zone. A security group, which is a virtual firewall that regulates incoming and outgoing traffic, protects the EC2 instance. A private key is kept on the local computer, while a public key is kept on the instance. To establish the user's identity, both keys are supplied as a key pair. The instance in this case is backed up using an Amazon EBS volume. An internet gateway connects the VPC to the internet.

2.2 Linux EC2 Instance

When you start an instance, the instance type you choose specifies the hardware of the host computer utilized for your instance. Each instance type has distinct computation, memory, and storage capabilities and is classified as an instance family based on these characteristics. Choose an instance type depending on the needs of the application or program you intend to run on your instance.

Amazon EC2 allots parts of the host computer's resources, such as CPU, memory, and instance storage, to a specific instance. Amazon EC2 uses the host computer's other resources, such as the network and disc subsystem, among other things.

3 DOCUMENTATION LINKS

1. Amazon EC2

<https://aws.amazon.com/ec2/>

2. Features of Amazon EC2

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/concepts.html>

3. Amazon EC2 FAQs

<https://aws.amazon.com/ec2/faqs/>

4. Getting Started with Amazon EC2 linux Instances

https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/EC2_GetStarted.html

4 PRE-REQUISITE

An AWS Account (Free or Paid).

Note: If you are using a Mac machine instead of Windows kindly refer to **Activity Guide (Lab): Connecting Linux EC2 Instance Using Mac** as we have create a separated guide and video for Mac users.

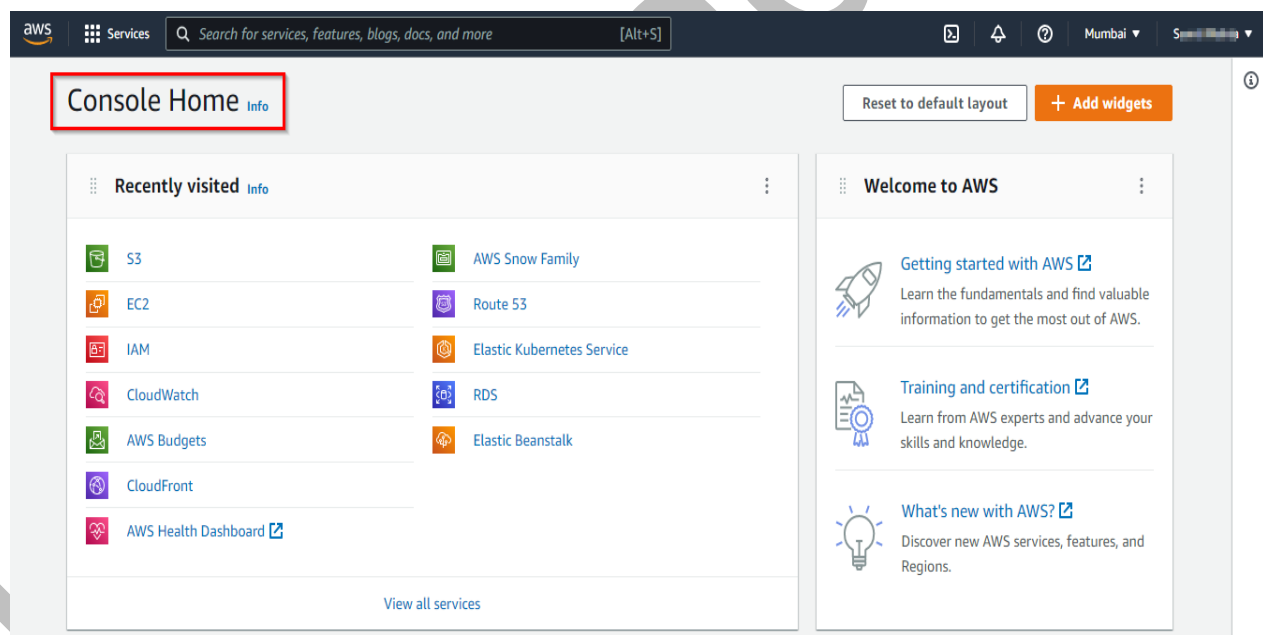
5 LAUNCHING A LINUX INSTANCE USING EC2

Launching a Linux Instance means you will create a Virtual Machine and access it using tools like Putty, MobaXterm etc. You shall be accessing this Virtual machine using internet, accessing on internet means these machines are physically present in the datacenters that lie in different Regions in the world. A datacenter is a place used to house computer systems and associated components, such as routers, switches, firewalls, storage systems, servers etc. So, you are basically accessing the machine present in such datacenters from anywhere in the world via internet. The launching of an instance does not depend on the underlying infrastructure whether your Laptop is having Windows or macOS.

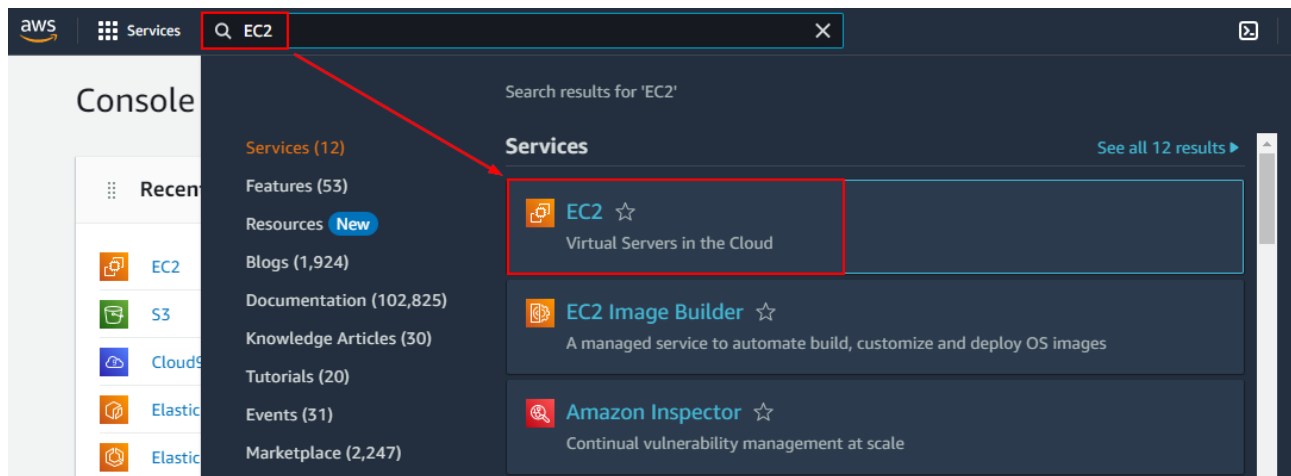
5.1 Creating an EC2 Instance

Note: Amazon EC2 basically provides a **Virtual Computing environment**, where they have pre-configured AMI (Amazon Machine Image) from which you can launch Virtual machines. In this section, we shall see the steps to create an configure the Linux EC2 instance from a pre-configured AMI named **Amazon Linux 2023 AMI**.

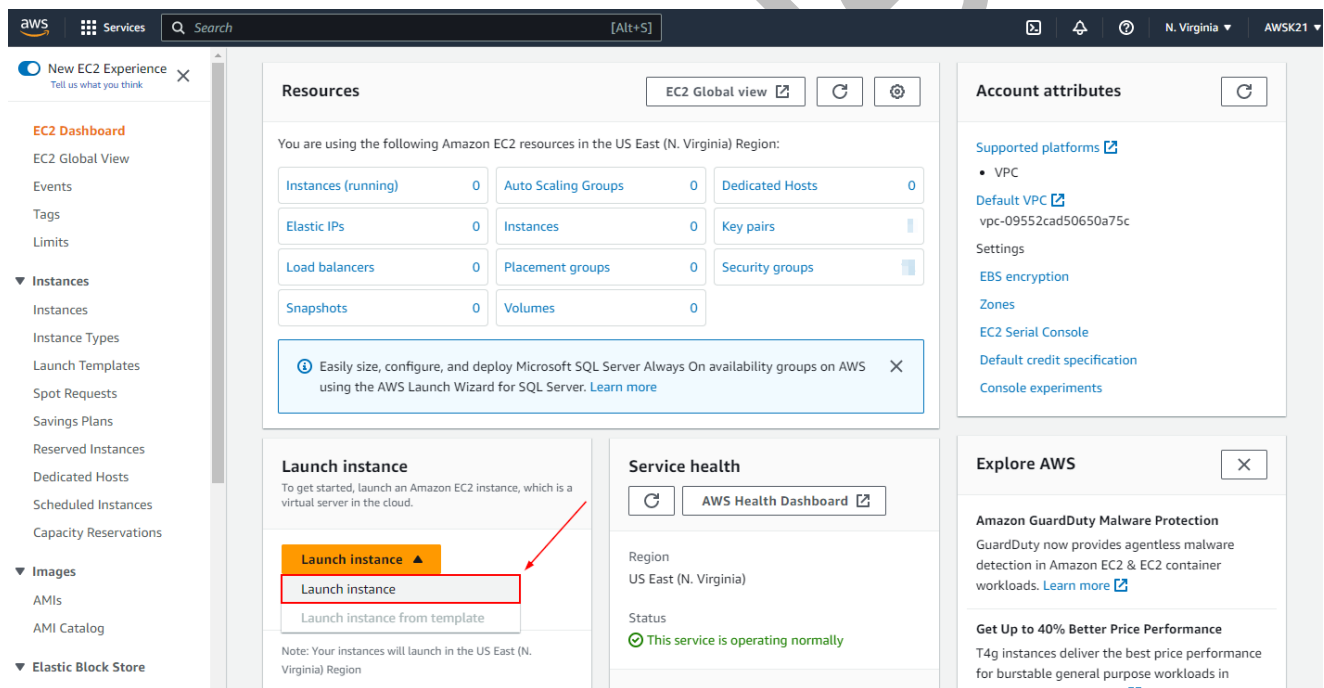
1. Log in to your AWS console.



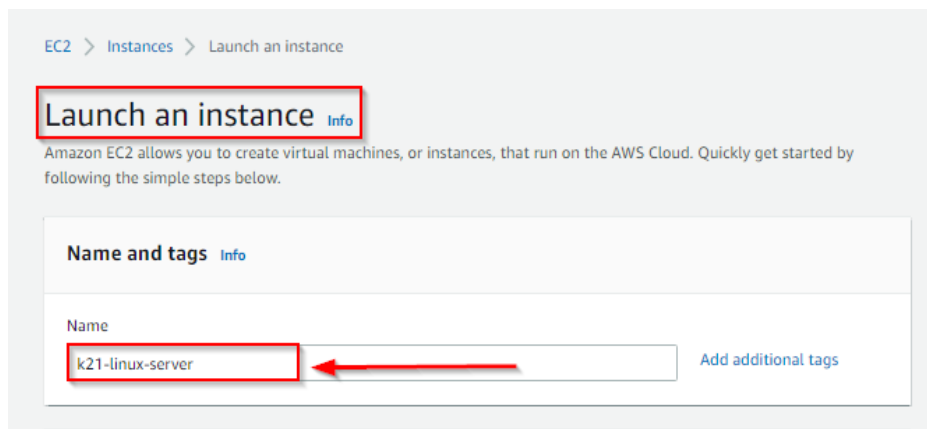
2. Search for **EC2** in the search bar and click on it.



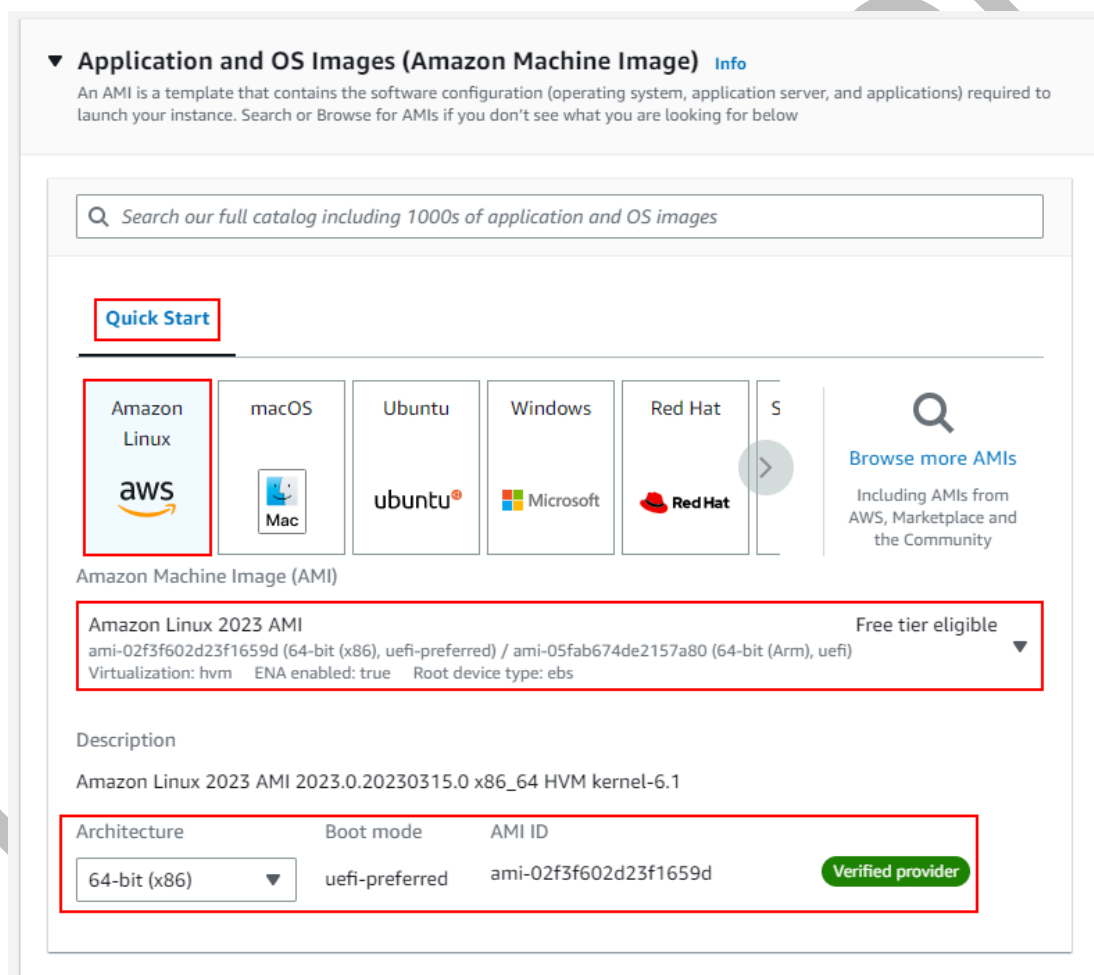
3. Once you are at the EC2 Dashboard, click on **Launch** to launch an instance.



4. In the Name and Tags step you can add tags to an instance, here tags help you to enable categorize AWS resources in different ways, for example, by owner, environment, or purpose.



5. Choose **Amazon Linux**, select **Amazon Linux 2023 AMI** architecture and Click on **Select**.



Note: Here, in the above step we are selecting the Amazon Linux 2023 AMI which is basically preconfigured with some tools installed on it like Amazon Linux 2023 AMI comes with AWS CLI pre-installed. AWS CLI is basically the Command Line Interface provided by AWS to access any of their services using CLI.

6. For Select the **t2.micro** instance type, if you want you may select another instance type but they are chargeable so we choose the t2.micro instance type which is eligible for the free tier and limited resources.

▼ Instance type [Info](#)

Instance type

t2.micro Free tier eligible
Family: t2 1 vCPU 1 GiB Memory
On-Demand Windows pricing: 0.0162 USD per Hour
On-Demand SUSE pricing: 0.0116 USD per Hour
On-Demand RHEL pricing: 0.0716 USD per Hour
On-Demand Linux pricing: 0.0116 USD per Hour

[Compare instance types](#)

selected key pair before you launch

[Create new key pair](#)

t1.micro Free tier eligible
Family: t1 1 vCPU 0.612 GiB Memory
On-Demand Linux pricing: 0.02 USD per Hour
On-Demand SUSE pricing: 0.02 USD per Hour
On-Demand RHEL pricing: 0.08 USD per Hour
On-Demand Windows pricing: 0.02 USD per Hour

t2.nano
Family: t2 1 vCPU 0.5 GiB Memory
On-Demand Linux pricing: 0.0058 USD per Hour
On-Demand SUSE pricing: 0.0058 USD per Hour
On-Demand Windows pricing: 0.0081 USD per Hour

t2.micro Free tier eligible
Family: t2 1 vCPU 1 GiB Memory
On-Demand Windows pricing: 0.0162 USD per Hour
On-Demand SUSE pricing: 0.0116 USD per Hour
On-Demand RHEL pricing: 0.0716 USD per Hour
On-Demand Linux pricing: 0.0116 USD per Hour

[Edit](#)

t2.small
Family: t2 1 vCPU 2 GiB Memory
On-Demand Windows pricing: 0.032 USD per Hour
On-Demand Linux pricing: 0.023 USD per Hour

Note: Here we must choose the instance type as **t2.micro**, because it is free tier eligible, i.e. you won't be charge for running the instance with this instance type as we need to perform the labs being in free tier. So, please make sure you select the t2.micro as the instance type else you shall be charged for running instances from Paid AMI's.

7. Select an existing key pair or create a new one, we will Create a new one, enter the name of the Key-pair as Linux-Key, select .ppk and Create the Key Pair.

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

Select [Create new key pair](#)

Create key pair

×

Key pairs allow you to connect to your instance securely.

Enter the name of the key pair below. When prompted, store the private key in a secure and accessible location on your computer. **You will need it later to connect to your instance.** [Learn more](#)

Key pair name

linux-key

The name can include upto 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ RSA
RSA encrypted private and public key pair

☐ ED25519
ED25519 encrypted private and public key pair (Not supported for Windows instances)

Private key file format

☐ .pem
For use with OpenSSH

☒ .ppk
For use with PuTTY

Cancel

Create key pair

8. Now select HTTP Port under Network Setting.

▼ Network settings

Info

Edit

Network

Info

vpc-09552cad50650a75c

Subnet

Info

No preference (Default subnet in any availability zone)

Auto-assign public IP

Info

Enable

Firewall (security groups)

Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

We'll create a new security group called 'launch-wizard-6' with the following rules:

☒ Allow SSH traffic from

Helps you connect to your instance

Anywhere

0.0.0.0/0

☐ Allow HTTPS traffic from the internet

To set up an endpoint, for example when creating a web server

☒ Allow HTTP traffic from the internet

To set up an endpoint, for example when creating a web server

⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

×

Note: We add http rule here because we need to access our Webpage. As the **communication** for request and response between Webserver and Browser **happens on port 80** which is **http**.

9. Now review all the things you have Configured and Click on **Launch Instance**.

Network settings Info Edit

Network Info
vpc-09552cad50650a75c

Subnet Info
No preference (Default subnet in any availability zone)

Auto-assign public IP Info
Enable

Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

We'll create a new security group called 'launch-wizard-6' with the following rules:

☒ Allow SSH traffic from
Helps you connect to your instance
Anywhere
0.0.0.0/0

☐ Allow HTTPS traffic from the internet
To set up an endpoint, for example when creating a web server

☒ Allow HTTP traffic from the internet
To set up an endpoint, for example when creating a web server

⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only. ✕

Summary

Number of instances Info
1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.0.2...read more
ami-02f3f602d23f1659d

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Free tier: In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in which t2.micro is unavailable) instance usage on free tier AMIs per month, 30 GiB of EBS storage, 2 million IOs, 1 GB of snapshots, and 100 GB of bandwidth to the internet. ✕

Cancel **Launch instance**

10. Linux Instance launched successfully.

EC2 > Instances > Launch an instance

Success
Successfully initiated launch of instance (i-0802130cd2ecab9fe)
▶ Launch log

Next Steps - preview

Q What would you like to do next with this instance, for example "create alarm" or "create backup"

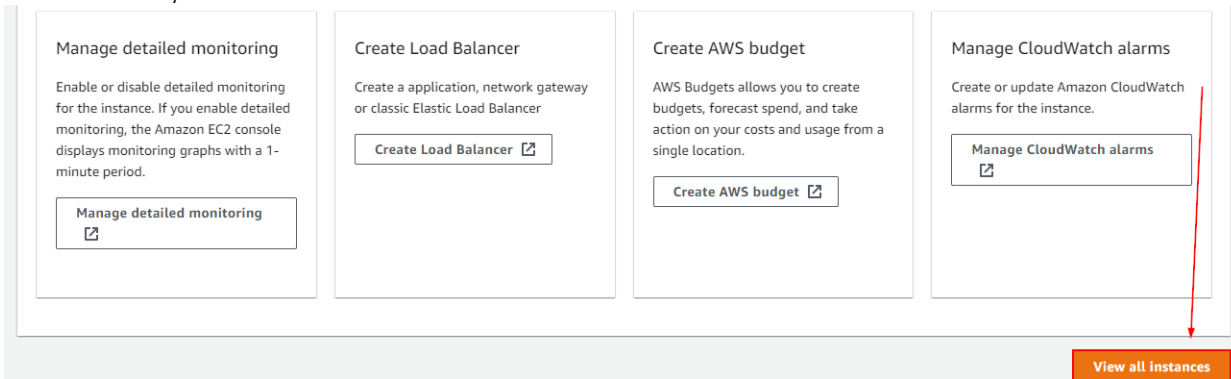
Create billing and free tier usage alerts
To manage costs and avoid surprise bills, set up email notifications for billing and free tier usage thresholds.
Create billing alerts

Connect to your instance
Once your instance is running, log into it from your local computer.
Connect to instance
Learn more

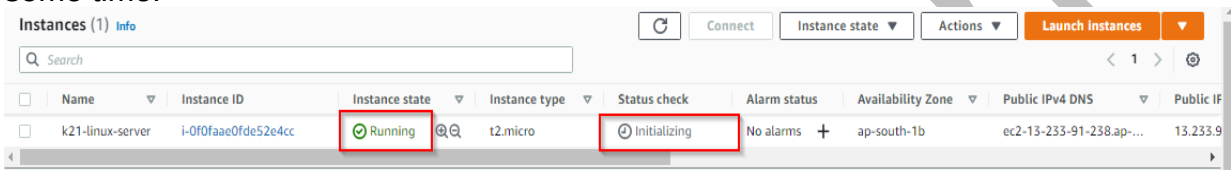
Connect an RDS database
Configure the connection between an EC2 instance and a database to allow traffic flow between them.
Connect an RDS database
Create a new RDS database
Learn more

Create EBS snapshot policy
Create a policy that automates the creation, retention, and deletion of EBS snapshots
Create EBS snapshot policy

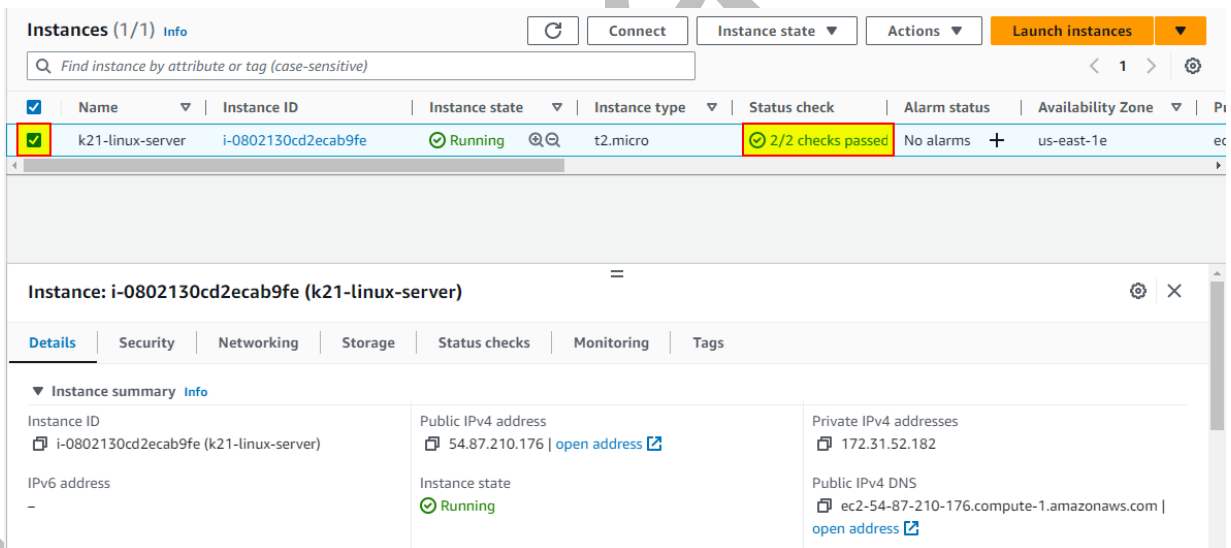
11. Scroll down, Click on **View all Instances**.



12. Here, you shall see your instance is launching and the **Status check** is **Initializing**, wait for some time.



13. Refresh and you shall see your instance are Up and Running, and **Status check** has changed to **2/2 checks**.



Now, we have successfully created our first Linux Instance using EC2.

Note: Now, wait for the Status check of the instance to change to 2/2 checks pass as the instance is launching from an AMI and it needs time to install all the preconfigured things like AWS CLI and other tools. So, please wait till the Status check becomes 2/2 checks pass and then only proceed with further steps, and if status check has not passed then you shall get error in the further steps so please wait.

5.2 Connecting to EC2 Linux Instance

Now we have launched the instance successfully, further we need to access the instance so, we have different ways for accessing an instance

If you are able to connect using one of the method you are good to proceed further. EC2 instance Direct connect is New, easy and most convenient to use. Putty & MobaXterm are optional way to connect to Linux EC2.

- a. Accessing via Browser using EC2 instance Connect.
- b. (Optional) Accessing using Putty (a Secure Shell SSH client)
- c. (Optional) Accessing using MobaXterm

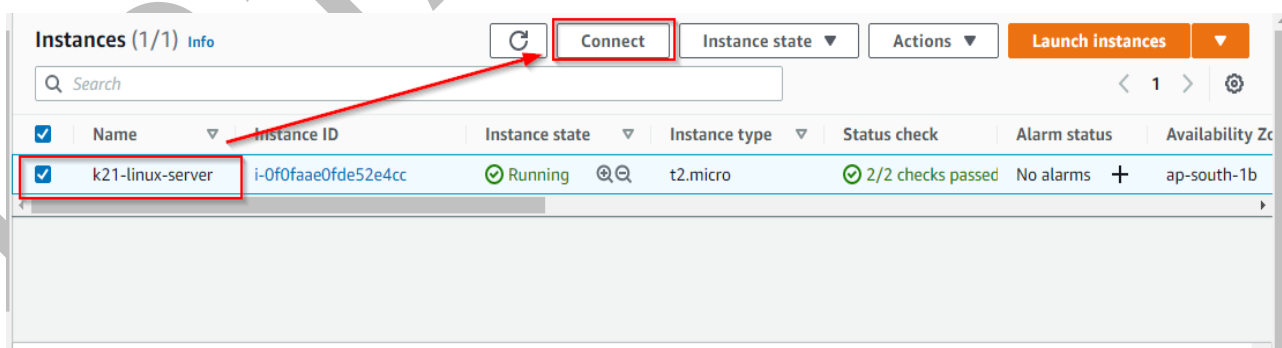
If you are beginner or new to IT kindly proceed with the first option (Via Browser) only and then come back to other options (b & c) after working on few other labs.

Note: If you hit any problem on connectivity please check out the troubleshooting section before reachout us on whatsapp or support@k21academy.com as your issue might be covered inside the troubleshooting section.

5.2.1 ACCESSING EC2 INSTANCE USING BROWSER SSH CONNECTION

Till now, we have launched an instance successfully, but we need to access it to run commands so for accessing our instance the easy way is you can access the CLI directly from the browser using EC2 instance connect that we shall see in further steps. This is the easier and most convenient way to connect to the EC2.

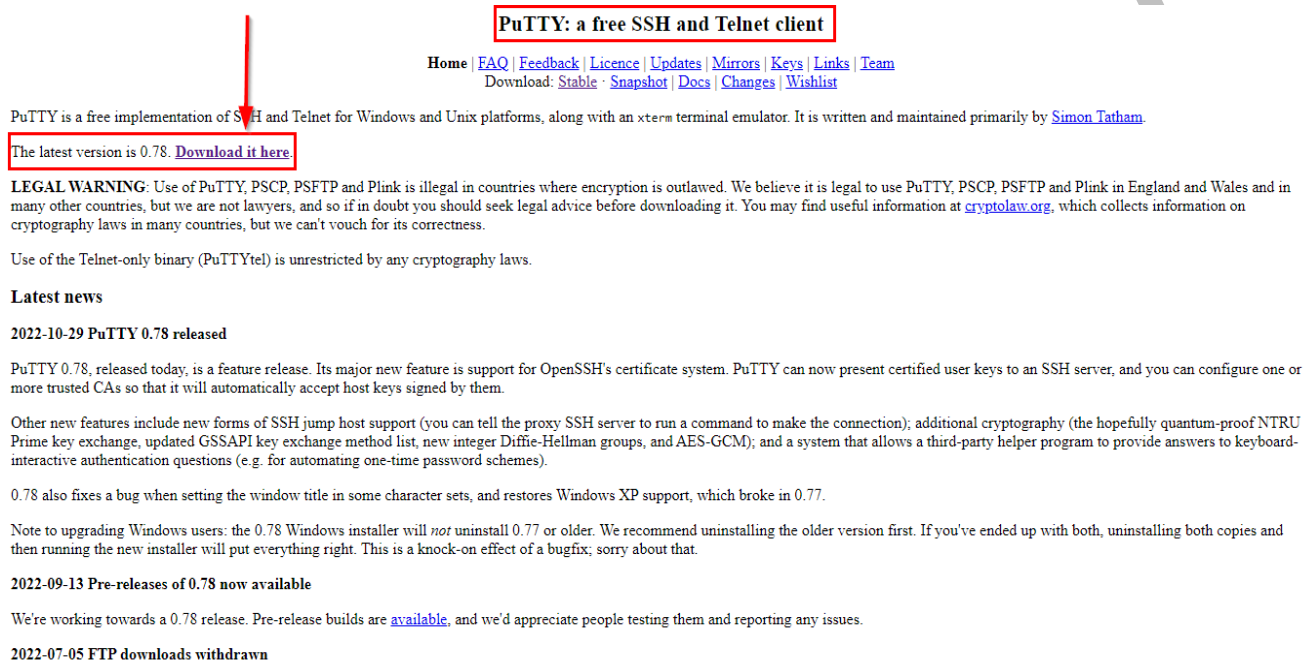
1. Select your Instance and Click on **Connect**.



5.2.2 (OPTIONAL) ACCESSING EC2 INSTANCE USING PUTTY

Note: PuTTY is an SSH and telnet client. PuTTY is a free and open-source terminal emulator, serial console and network file transfer application. It supports several network protocols, including SCP, SSH, Telnet, rlogin, and raw socket connection. In this section, we will **download and install Putty** and will try to SSH (Secure Shell) into our instance using it.

1. Download and Install Putty from <http://www.chiark.greenend.org.uk/~sgtatham/putty/> Once you Click on the link you shall see this screen, Click on **Download it here**.



PuTTY: a free SSH and Telnet client

Home | [FAQ](#) | [Feedback](#) | [Licence](#) | [Updates](#) | [Mirrors](#) | [Keys](#) | [Links](#) | [Team](#)
Download: [Stable](#) · [Snapshot](#) | [Docs](#) | [Changes](#) | [Wishlist](#)

PuTTY is a free implementation of SSH and Telnet for Windows and Unix platforms, along with an xterm terminal emulator. It is written and maintained primarily by [Simon Tatham](#).

The latest version is 0.78. [Download it here](#).

LEGAL WARNING: Use of PuTTY, PSCP, PSFTP and Plink is illegal in countries where encryption is outlawed. We believe it is legal to use PuTTY, PSCP, PSFTP and Plink in England and Wales and in many other countries, but we are not lawyers, and so if in doubt you should seek legal advice before downloading it. You may find useful information at cryptolaw.org, which collects information on cryptography laws in many countries, but we can't vouch for its correctness.

Use of the Telnet-only binary (PuTTYtel) is unrestricted by any cryptography laws.

Latest news

2022-10-29 PuTTY 0.78 released

PuTTY 0.78, released today, is a feature release. Its major new feature is support for OpenSSH's certificate system. PuTTY can now present certified user keys to an SSH server, and you can configure one or more trusted CAs so that it will automatically accept host keys signed by them.

Other new features include new forms of SSH jump host support (you can tell the proxy SSH server to run a command to make the connection); additional cryptography (the hopefully quantum-proof NTRU Prime key exchange, updated GSSAPI key exchange method list, new integer Diffie-Hellman groups, and AES-GCM); and a system that allows a third-party helper program to provide answers to keyboard-interactive authentication questions (e.g. for automating one-time password schemes).

0.78 also fixes a bug when setting the window title in some character sets, and restores Windows XP support, which broke in 0.77.

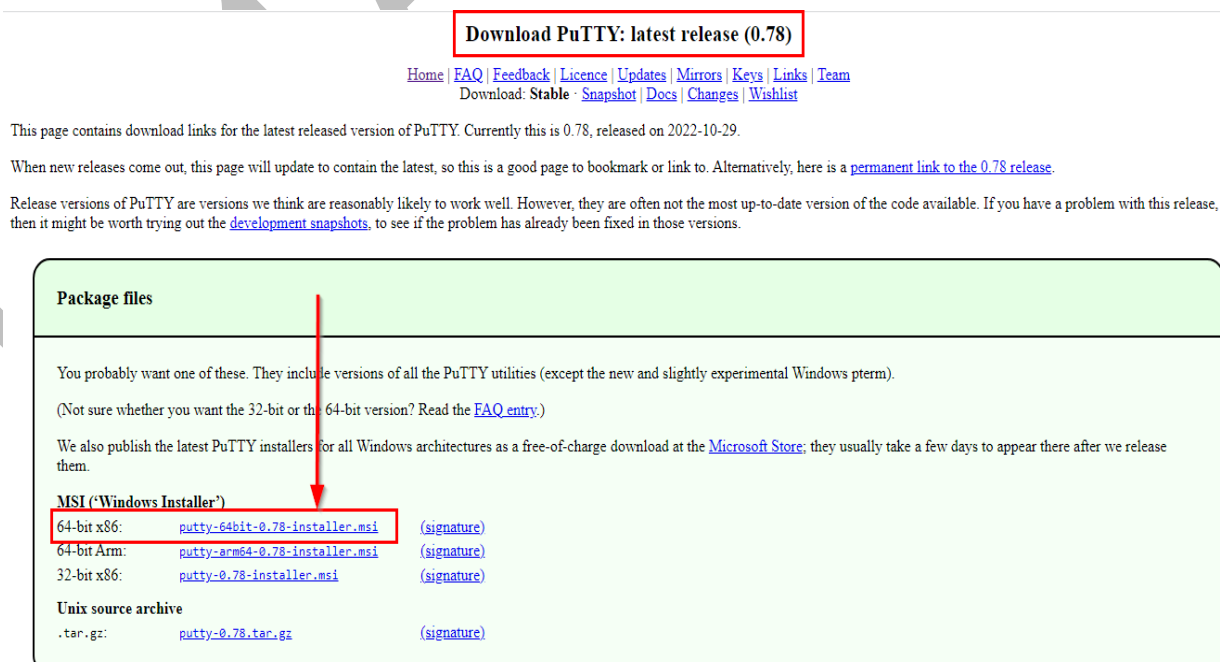
Note to upgrading Windows users: the 0.78 Windows installer will *not* uninstall 0.77 or older. We recommend uninstalling the older version first. If you've ended up with both, uninstalling both copies and then running the new installer will put everything right. This is a knock-on effect of a bugfix; sorry about that.

2022-09-13 Pre-releases of 0.78 now available

We're working towards a 0.78 release. Pre-release builds are [available](#), and we'd appreciate people testing them and reporting any issues.

2022-07-05 FTP downloads withdrawn

2. Now Click on **64-bit x86** and install it once downloaded.



Download PuTTY: latest release (0.78)

Home | [FAQ](#) | [Feedback](#) | [Licence](#) | [Updates](#) | [Mirrors](#) | [Keys](#) | [Links](#) | [Team](#)
Download: [Stable](#) · [Snapshot](#) | [Docs](#) | [Changes](#) | [Wishlist](#)

This page contains download links for the latest released version of PuTTY. Currently this is 0.78, released on 2022-10-29.

When new releases come out, this page will update to contain the latest, so this is a good page to bookmark or link to. Alternatively, here is a [permanent link to the 0.78 release](#).

Release versions of PuTTY are versions we think are reasonably likely to work well. However, they are often not the most up-to-date version of the code available. If you have a problem with this release, then it might be worth trying out the [development snapshots](#), to see if the problem has already been fixed in those versions.

Package files

You probably want one of these. They include versions of all the PuTTY utilities (except the new and slightly experimental Windows pterm).
(Not sure whether you want the 32-bit or the 64-bit version? Read the [FAQ entry](#).)

We also publish the latest PuTTY installers for all Windows architectures as a free-of-charge download at the [Microsoft Store](#); they usually take a few days to appear there after we release them.

MSI ('Windows Installer')

64-bit x86:	putty-64bit-0.78-installer.msi	(signature)
64-bit Arm:	putty-arm64-0.78-installer.msi	(signature)
32-bit x86:	putty-0.78-installer.msi	(signature)

Unix source archive

.tar.gz:	putty-0.78.tar.gz	(signature)
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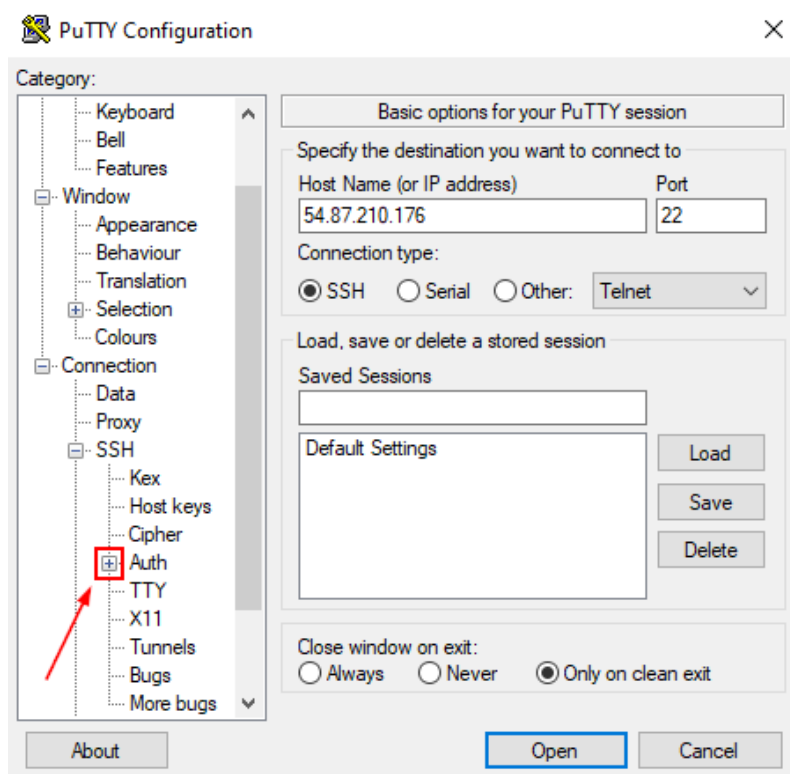
- Now Select your instance and Copy the **public IPv4 address** under Details as shown below.

The screenshot shows the AWS Management Console 'Instances' page. A table lists instances, with 'k21-linux-server' (ID: i-0802130cd2ecab9fe) highlighted in a red box. Below the table, the 'Details' tab for this instance is open. In the 'Instance summary' section, the 'Public IPv4 address' is '54.87.210.176', which is also highlighted with a red box and a red arrow pointing to it. Other details visible include the instance state 'Running' and various DNS and private IP addresses.

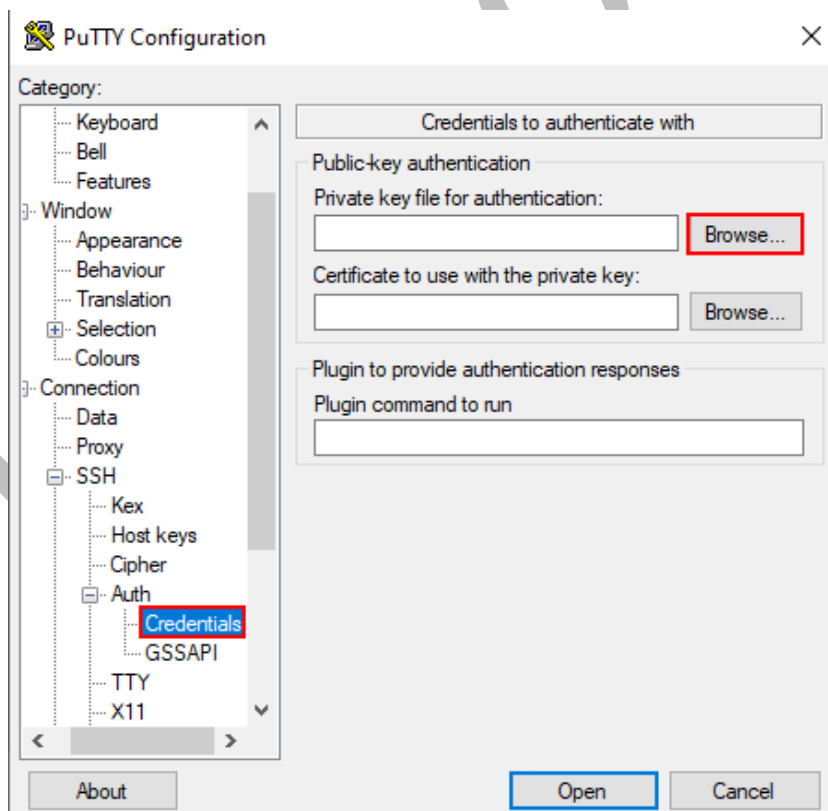
- Now Open **PuTTY** and paste the Public-ip address under Host Name (or IP address) you copied and click on **plus mark on SSH**.

The screenshot shows the 'PuTTY Configuration' dialog box. In the left-hand 'Category' tree, the 'SSH' option under the 'Connection' category is selected and highlighted with a red box and a red arrow. On the right, the 'Basic options for your PuTTY session' are visible. The 'Host Name (or IP address)' field contains '54.87.210.176' and is also highlighted with a red box. The 'Port' is set to '22' and the 'Connection type' is set to 'SSH'.

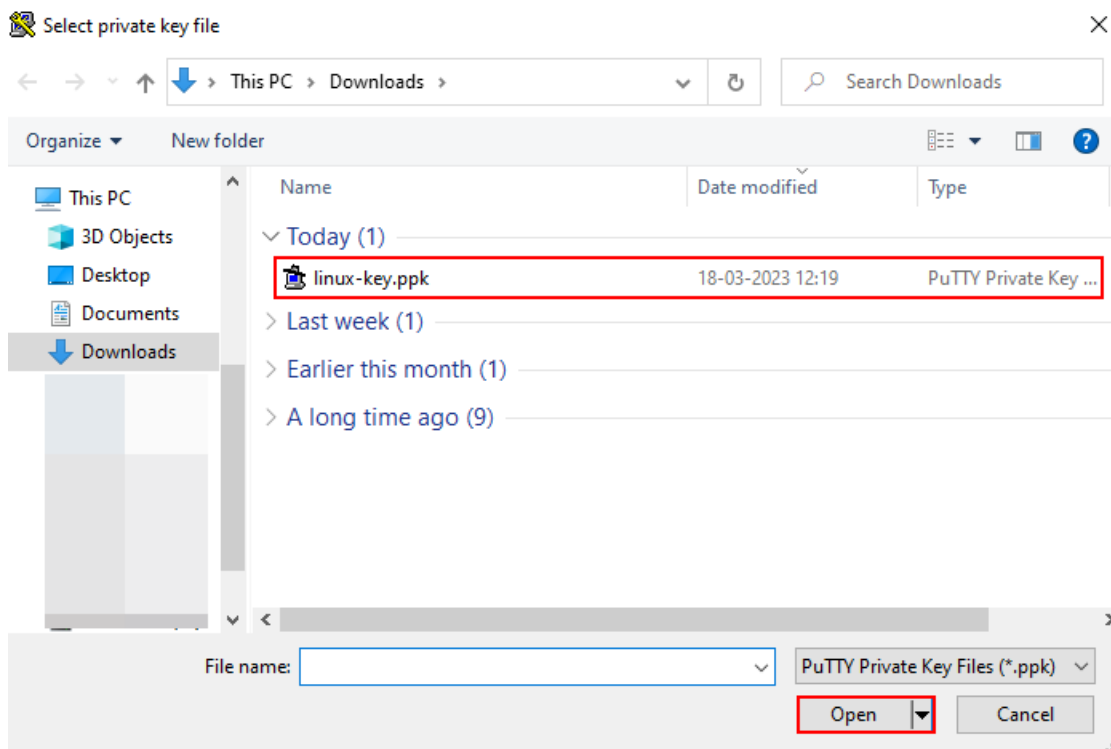
5. Under it now Click on “+” and expand **Auth**.



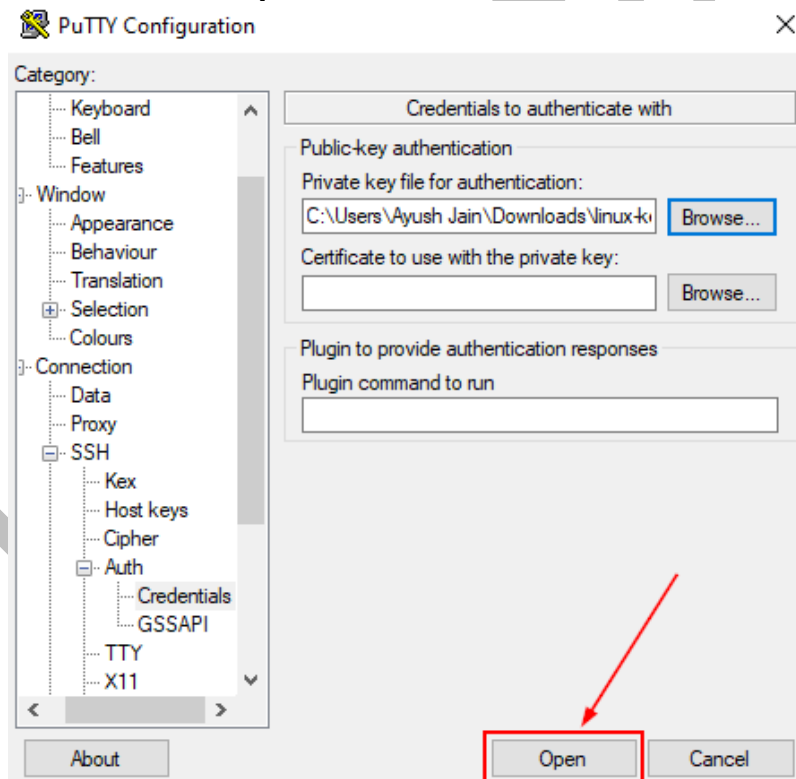
6. Now Click on **Credential** and Browse the **Private key file**.



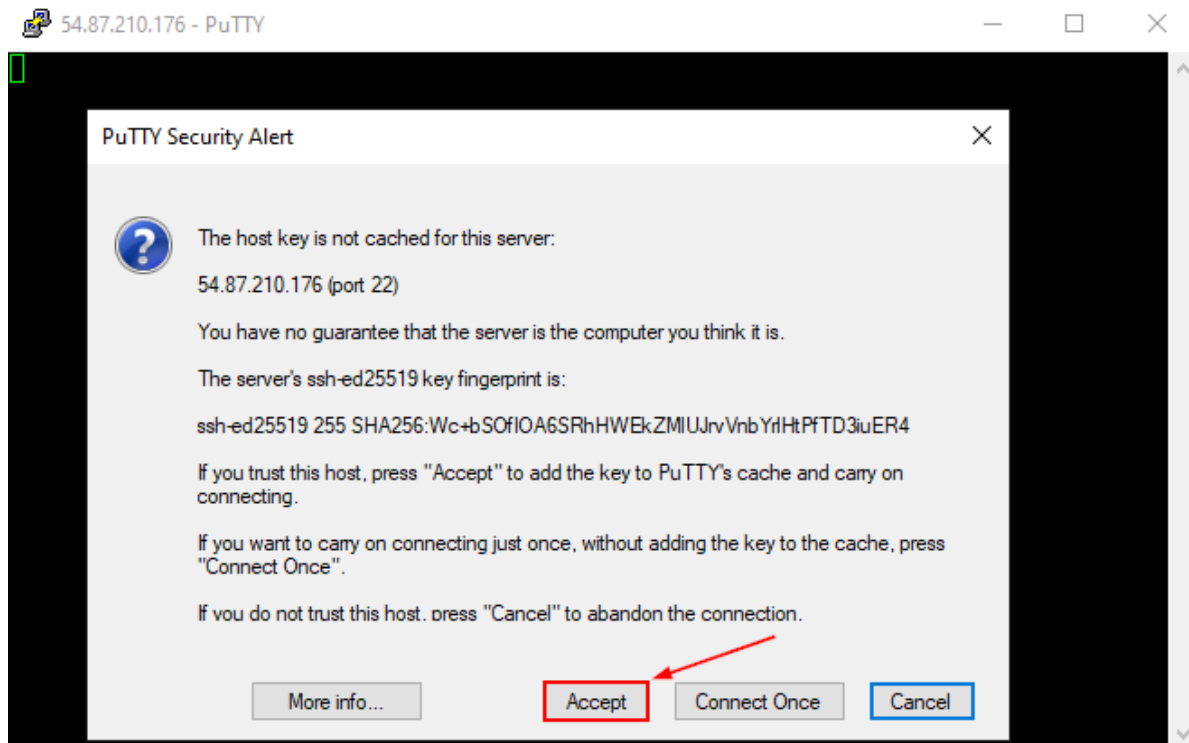
7. Now Select our **.ppk file** and Click on **Open**.



8. Now click n the **Open**.



9. Now Click on **Accept**.



10. Now type **ec2-user** for login as and press Enter. Now you have successfully logged to AWS Linux EC2 Instance.



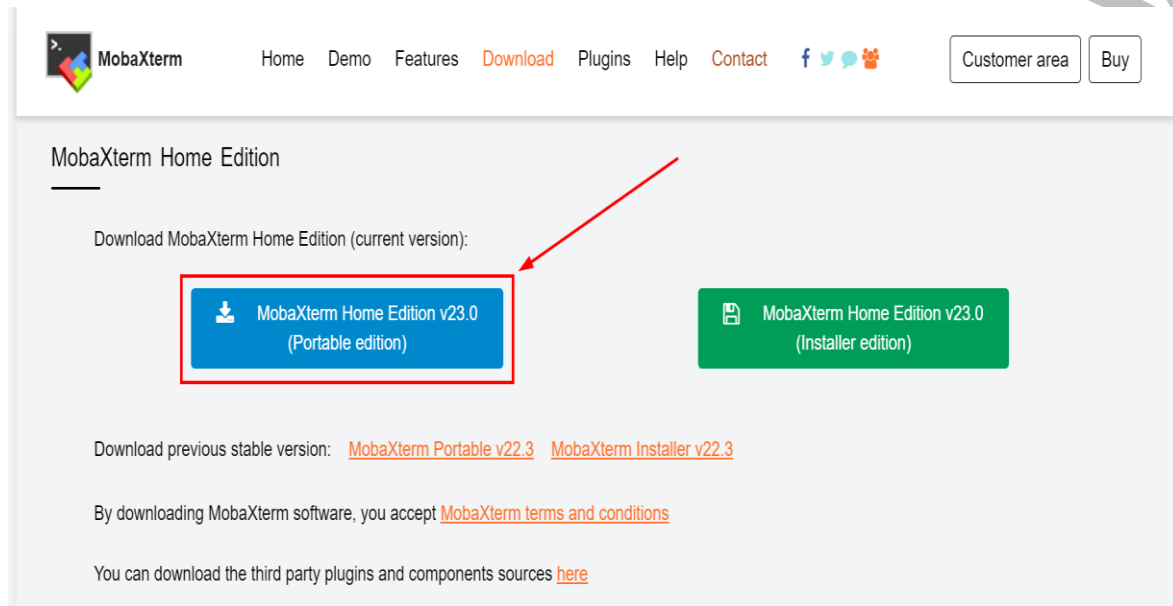
Note: Don't forget to type **ec2-user** to login as it is the default user by which you login into a Linux EC2 instance.

5.2.3 (OPTIONAL) ACCESSING EC2 INSTANCE USING MOBAXTERM

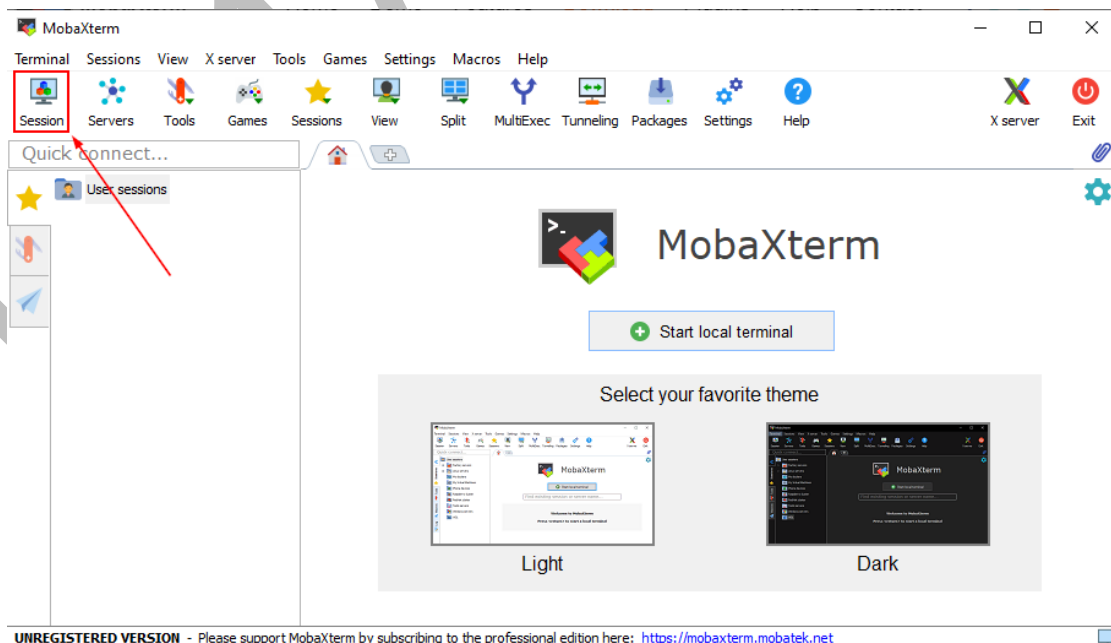
MobaXterm provides all the important remote network tools (SSH, RDP, X11, SFTP, FTP, Telnet, Rlogin) to Windows desktop, in a single portable exe file which works out of the box. Some plugins can be **used** to add functions to **MobaXterm** such as Unix commands (bash, ls, cat, sed, grep, awk, rsync).

Download link for MobaXterm (Windows):

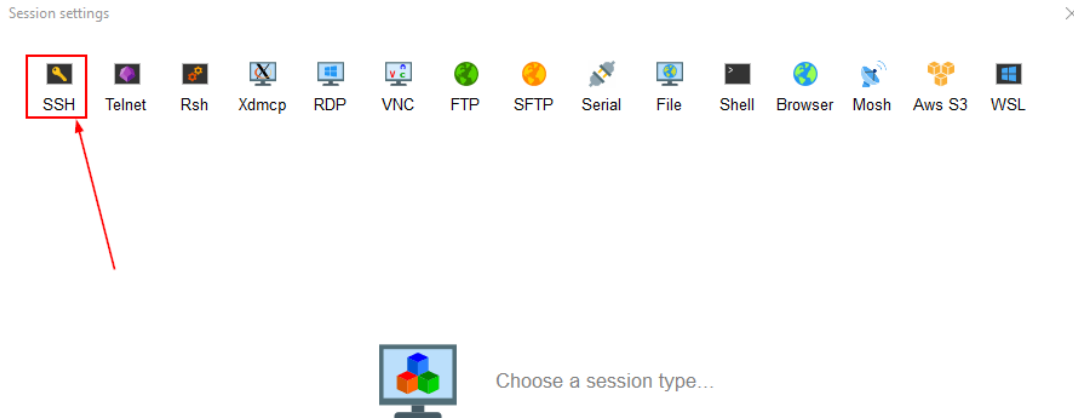
<https://mobaxterm.mobatek.net/download-home-edition.html>



1. Open **MobaXterm** and click on **Session**

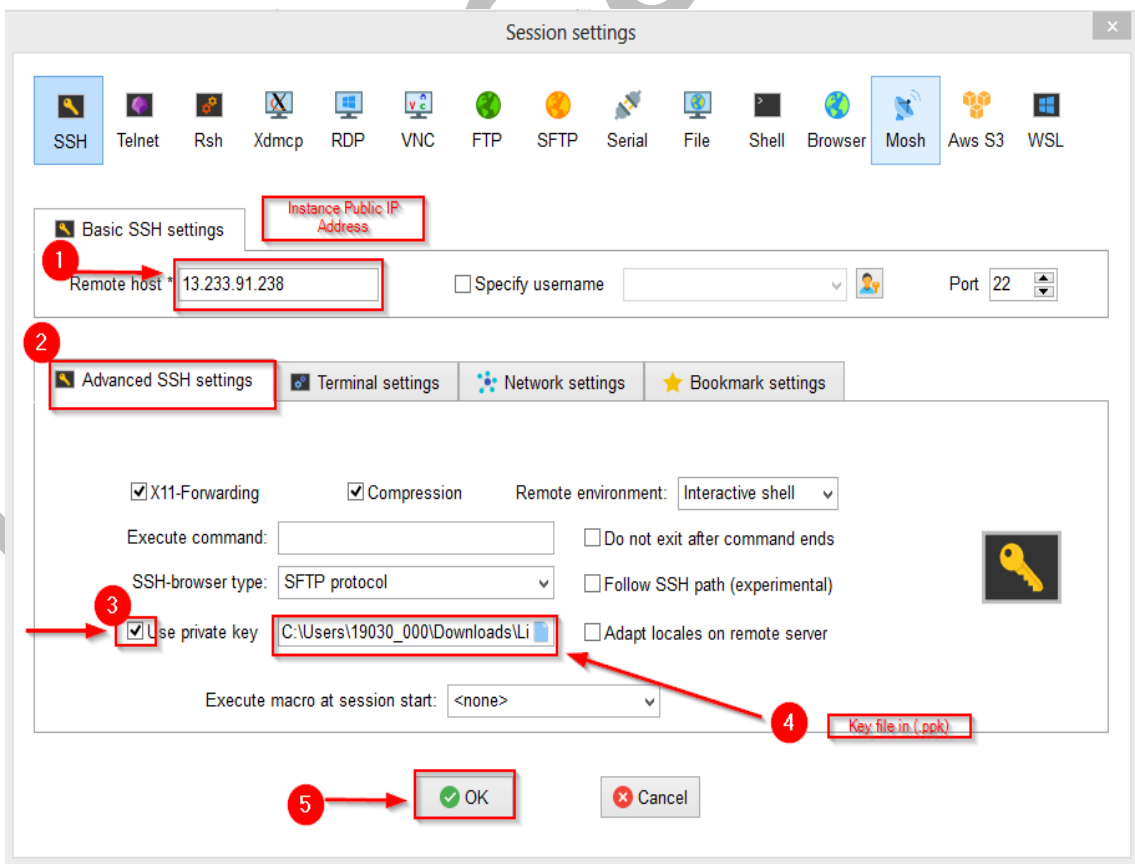


2. A dialog box will open and click on **SSH** at the top right corner.

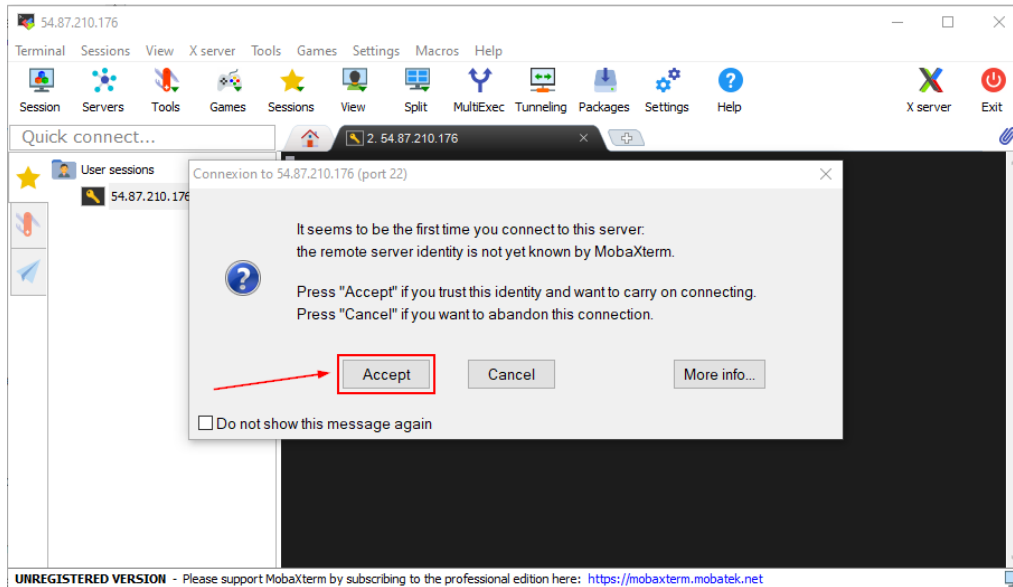


3. Now configure basic **SSH Settings**

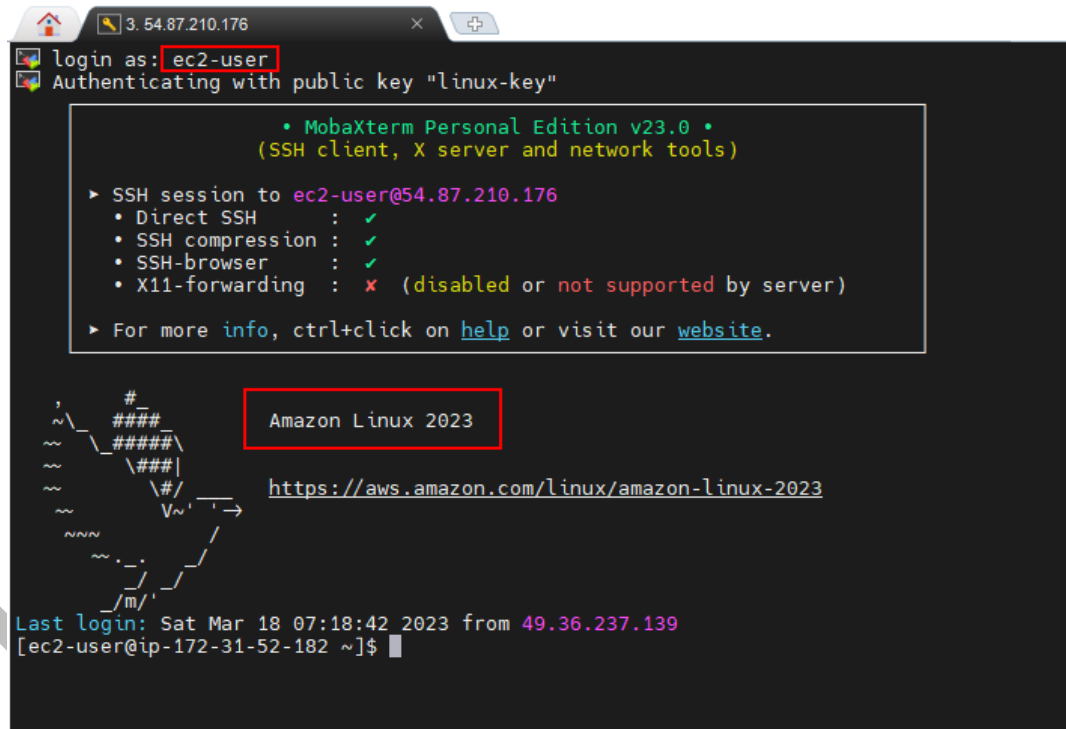
- In the **Remote host** : Type or Paste the **Public IP** of Linux Instance
- Click on Advanced SSH settings
- Click on **Use private key** checkbox and provide your **private (ppk)**
- Finally click on **OK**



4. Click on **Accept**.



5. Type `ec2-user` and press enter. Now you have successfully logged in to AWS Linux EC2 Instance



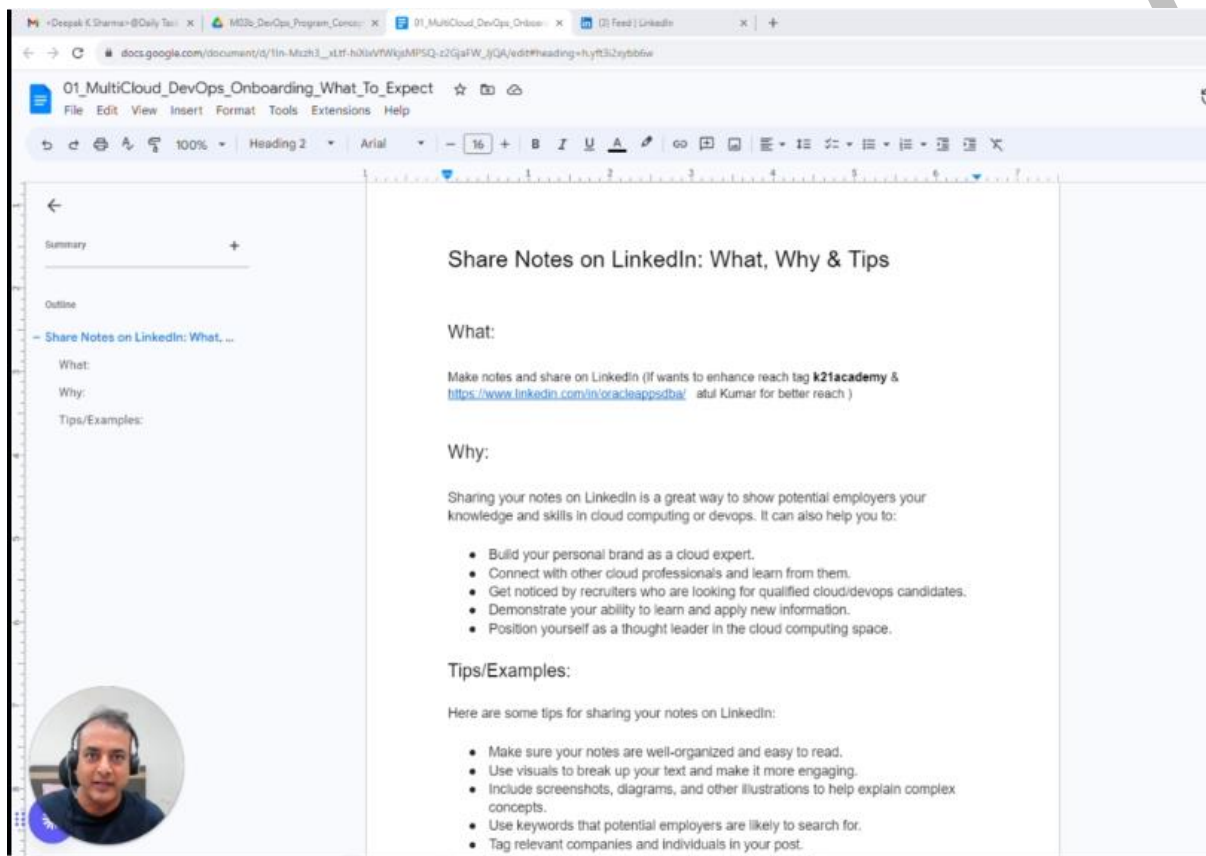
Thus we have successfully logged in to the Linux Instance using MobaXterm

Note: If you are not able to connect at this point kindly refer to the troubleshooting section 8.1

6 SHARE YOUR LEARNINGS

In this section, you are going to share whatever you learned during this lab on LinkedIn and WhatsApp

 <https://www.loom.com/share/d91f9abbec794f4f85e7d99bb82c2ff6>



- As you know, building a strong professional profile is essential in today's job market. One way to showcase your skills and knowledge is by sharing your labs and projects on LinkedIn.
- Your LinkedIn profile is a powerful tool in your job search. Sharing your labs and projects is just one way to demonstrate your expertise and stand out to potential employers. If you don't have LinkedIn, we strongly recommend you create one for yourself.
- By doing so, you can demonstrate to potential employers or connections that you have hands-on experience in your field and are actively engaged in learning and growing your skills.

6.1 On LinkedIn

Take a screenshot of lab and share it on your LinkedIn. So, this will attract recruiters and employers on your profile and increase reach. Do remember to tag **K21Academy**, as we will circulate in our network too, to increase your reach.

Please refer these following examples on how to share on LinkedIn (don't limit yourself to just these type). Share learnings in points, notes 📝 or anything that can help others.

Here is a sample which includes steps inside a post followed by screenshot:

Sumti Mehta • 1st
Cloud Computing | Amazon Web Services | 1x AWS Certified | Technical Support...
2w • Edited • 🌐

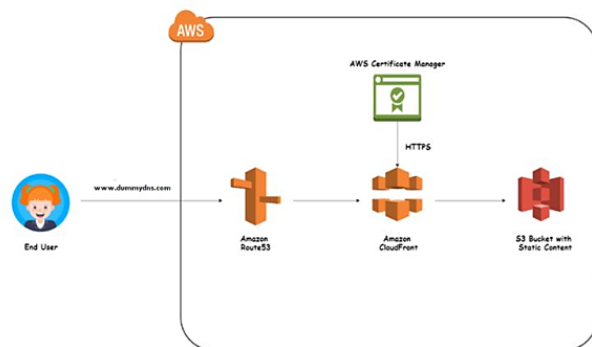
Hi There,

Today I have Provisioned an SSL/TLS Certificates using AWS Certificate Manager. It was fun and learned a lot of services integrated in the deployment.

To do this deployment I had to gone through the following steps:

1. Created an S3 bucket and enabled Static Website Hosting
2. Created a Hosted Zone and Mapped DNS Using Route53
3. Requested SSL/TLS certificates using AWS Certificate Manager
4. And finally, Configured CloudFront Distribution

Thanks, [K21Academy: Learn Cloud From Experts](#) For providing me with the required resources. [#aws](#) [#awscloud](#) [#awsservices](#) [#learning](#) [#cloud](#) [#cloud](#) [#project](#) [#learning](#)



Sumti Mehta and 15 others

Sahid Anwar Hussain • 1st
DevOps Engineer & Cloud Support at K21Academy: Learn Cloud From Experts
2w • 🌐

Hi There,

Today I have deployed an awesome tool to monitor an entire K8s cluster. It was fun and learned a lot during the deployment.

To do this deployment I had to gone through the following steps:

1. Created an multinode K8s cluster.
2. Created a namespace named "monitoring".
3. Created clusterRole & clusterRole Binding for Prometheus.
4. Created the ConfigMap including all the necessary Data for deploying Prometheus
5. Created a Deployment file for Prometheus
6. Finally I ran all the files and wallah.. Its done.

Thanks, [K21Academy: Learn Cloud From Experts](#) For providing me with the required resources. [#kubernetes](#) [#k8s](#) [#devops](#) [#learning](#) [#cloud](#) [#cloud](#) [#project](#) [#learning](#) [#prometheus](#) [#monitoring](#)

The screenshot shows the Prometheus 'Targets' page. It lists various endpoints and their status. The 'kube-state-metrics' endpoint is highlighted, showing it is 'UP' and has been scraped successfully. Other endpoints like 'kubernetes-apiversers' and 'kubernetes-cadvisor' are also listed with their respective statuses and scrape details.

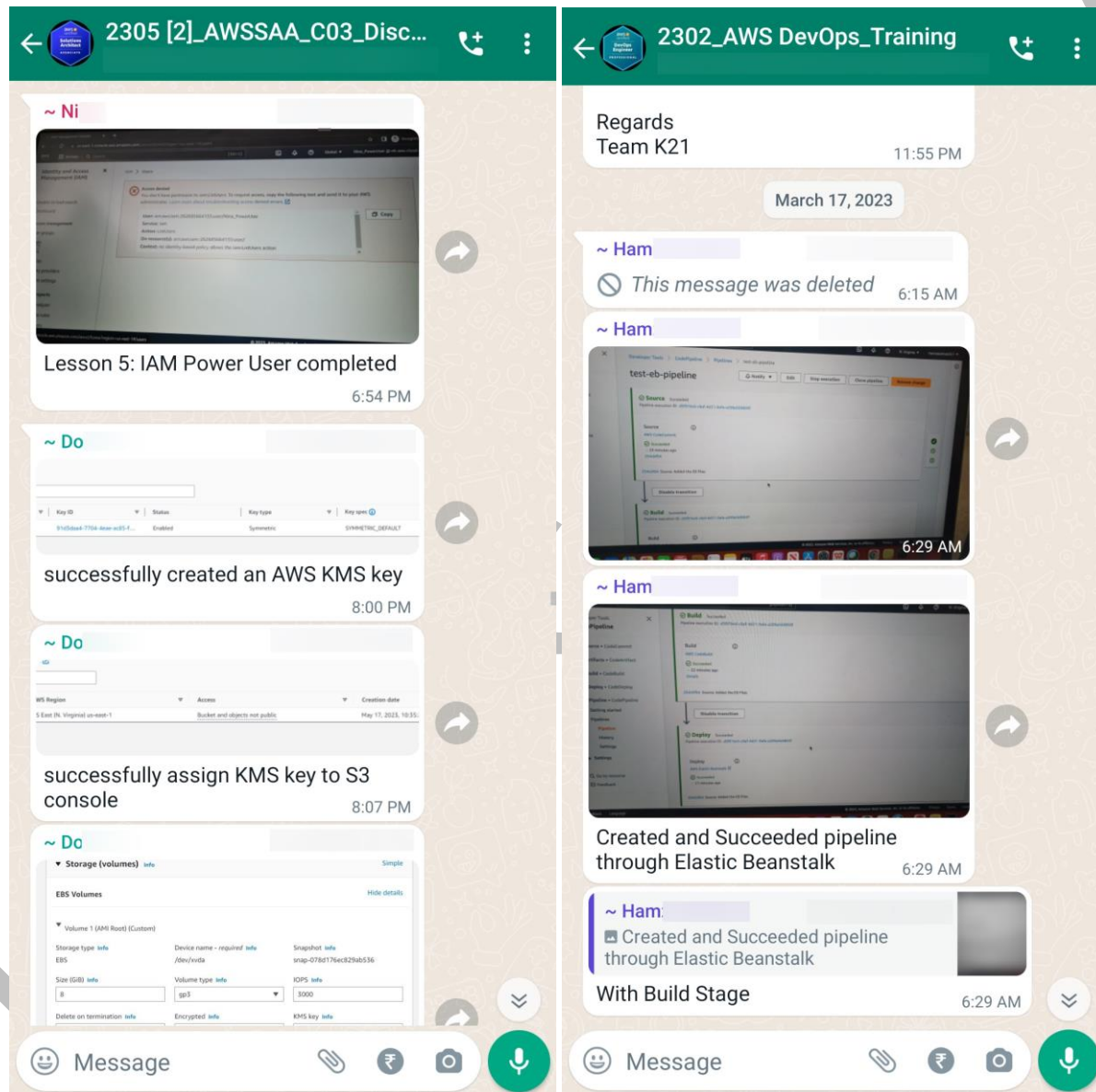
Endpoint	Status	Labels	Last Scrape	Scrape Duration	Error
kube-state-metrics (0/1 up)	UP	kubernetes="k8s" kube-state-metrics="true" kube-system="true"	3.999s ago	12.622ms	
kubernetes-apiversers (1/1 up)	UP	kubernetes="k8s" kube-system="true"	1.993s ago	95.479ms	
kubernetes-cadvisor (3/3 up)	UP	kubernetes="k8s" kube-system="true"	3.493s ago	38.210ms	

Suraj . and 12 others

6.2 On WhatsApp

Share screenshot of lab in WhatsApp group so this will encourage others and also boost your confidence on the progressive path you are following.

Also, we will keep track of your progress on labs so do share your screenshot in WhatsApp group too.



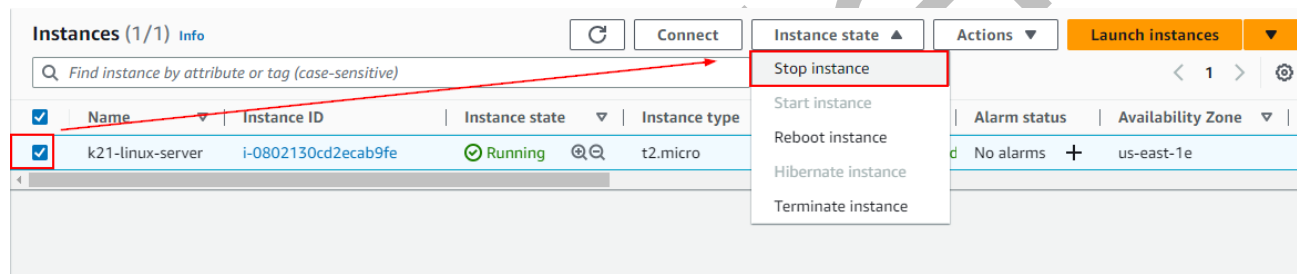
7 DELETING/CLEANUP

In this section we will be deleting all the resources we have used for this lab in order to be in free tier limit, as keeping these resources running will be chargeable.

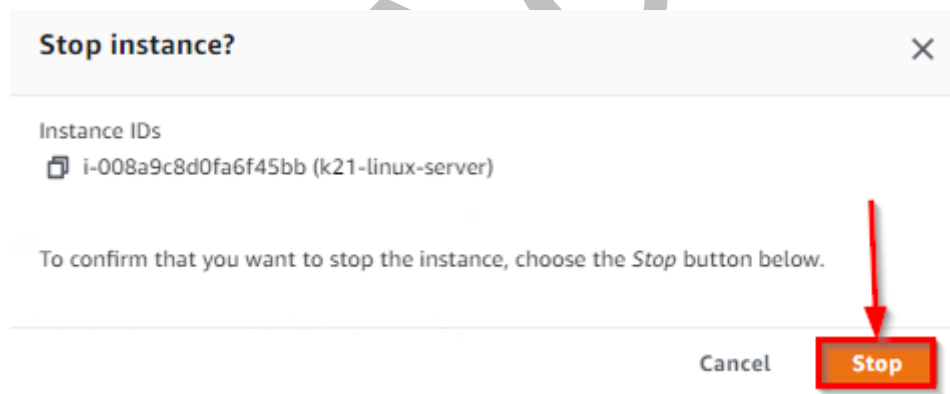
7.1 Stopping the Linux Instance

Note: We would always recommend to Terminate the instance if there is no requirement of it. But many times, it may happen that you shall require the instance after some time to continue with your work, then you shall stop the instance. Stopping the instance will not charge you but you shall be charged for the volume/storage attached to it. So, the charges shall be minimum as compared to running instance, but you will still get charged.

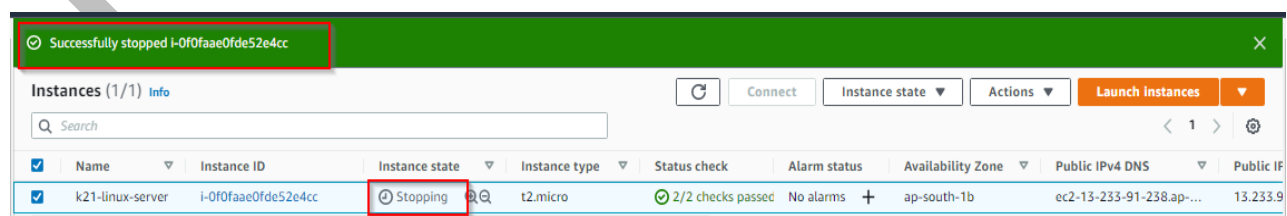
1. Back on EC2 Console, select our Instance, under **instance state** Click on **Stop instance**.



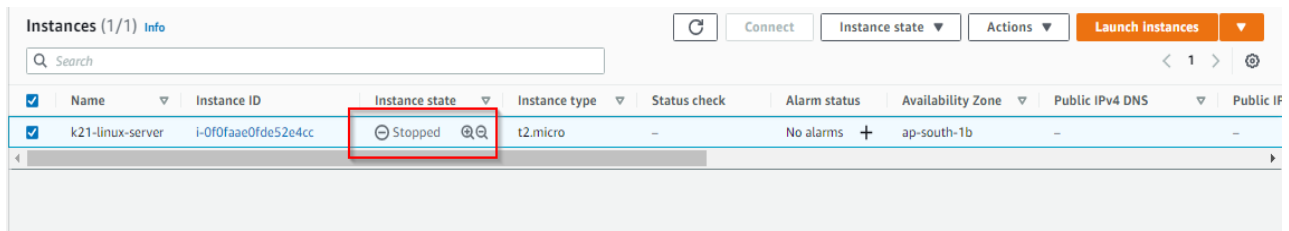
2. Click on **Stop** to **Confirm**.



3. Now you shall see our instance is in stopping state, wait for some time.



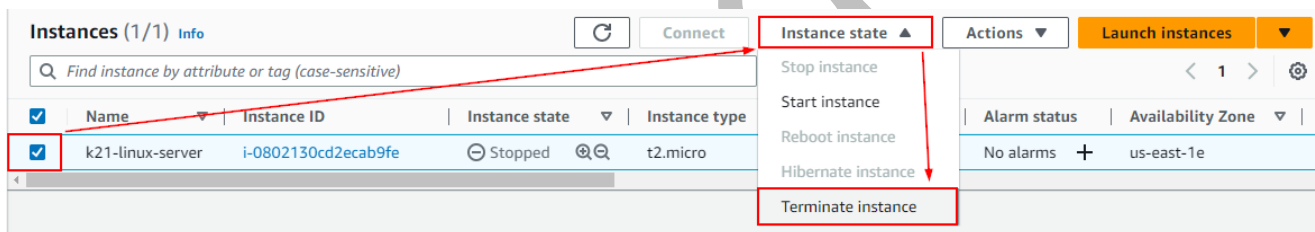
- Refresh and you shall see our instance have stopped.



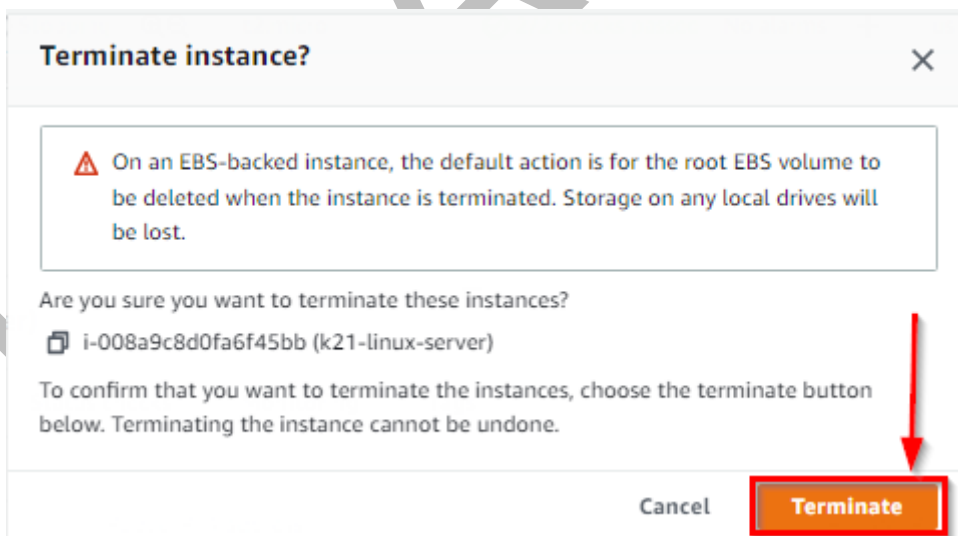
7.2 Terminating the Linux Instance

Note: In this section, we shall see the steps to Terminate the EC2 instance, as if the instances are not terminated once there is no requirement of it then you shall be charged for using those resources. So, to be in free tier and complete all the labs please terminate the resources here the EC2 instance when in no use.

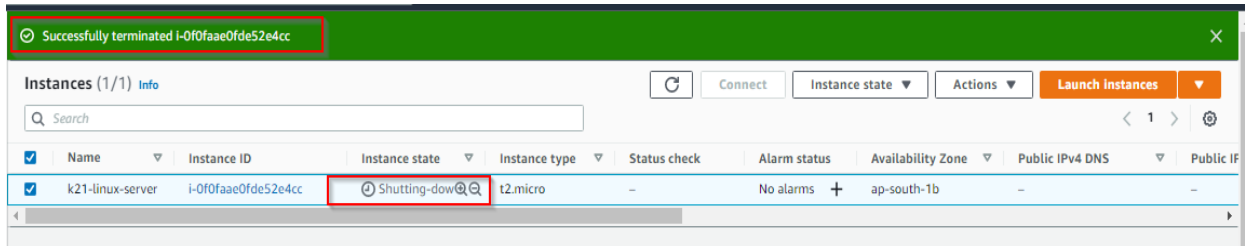
- Select our instance, under **Instance state** Click on **Terminate instance**.



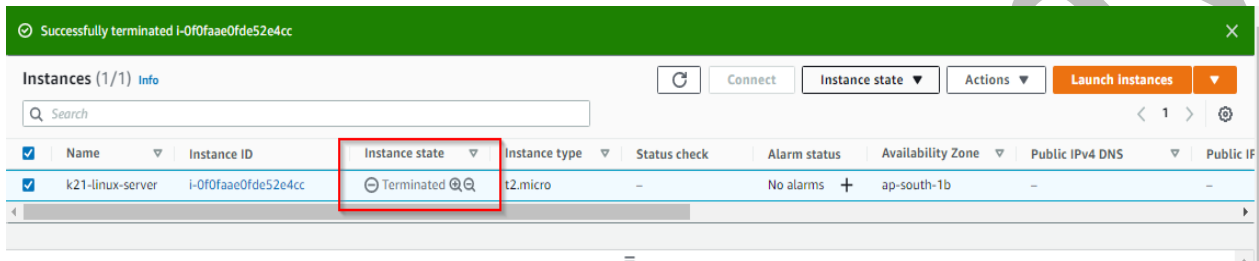
- Now Click on **Terminate** to Confirm.



3. Your instance is in **shutting down** state, wait for some time.



4. Refresh and you shall see our instance is **Terminated**.



Thus, we have successfully Terminated the Linux EC2 instance.

8 TROUBLESHOOTING

In this Section we will be adding some common error you may encounter while performing this lab.

8.1 Authentication Method Not Supported

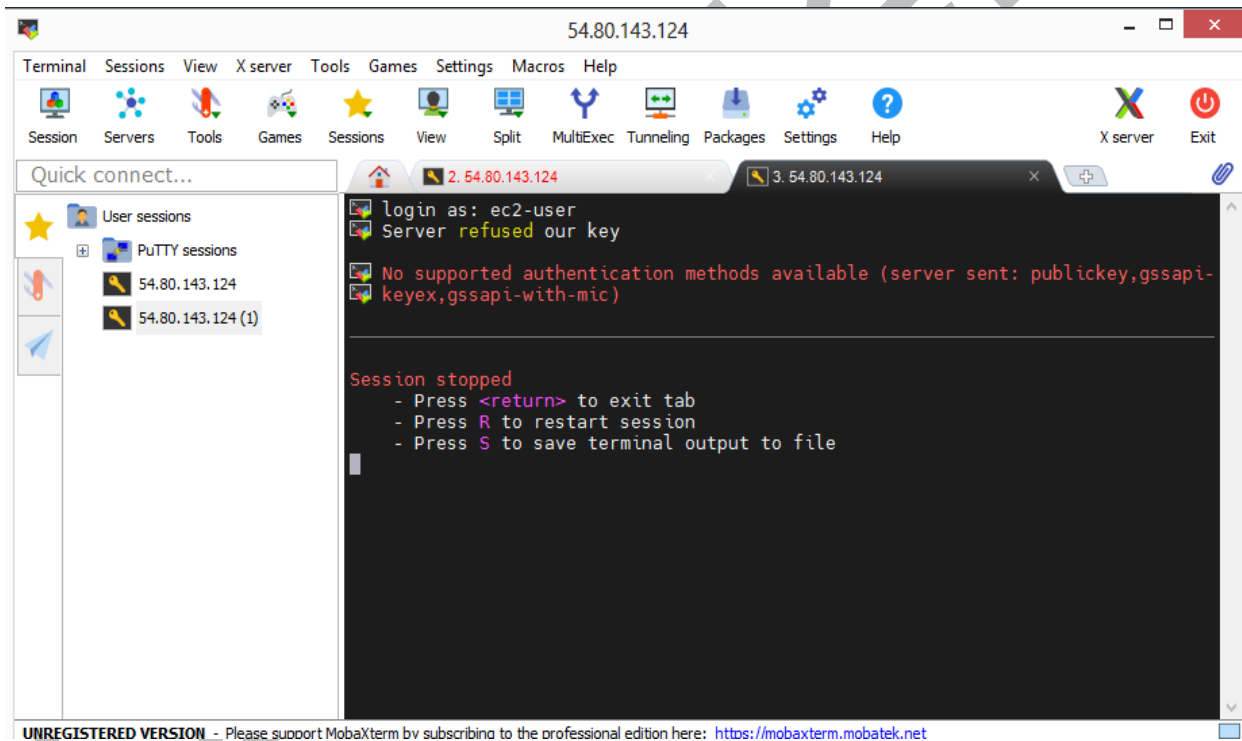
Sometimes you can get this error message while using MobaXterm.

8.1.1 ERROR

Permission denied (publickey,gssapi-keyex,gssapi-with-mic).

8.1.2 CAUSE

The SSH port may not be opened.



8.1.3 FIX

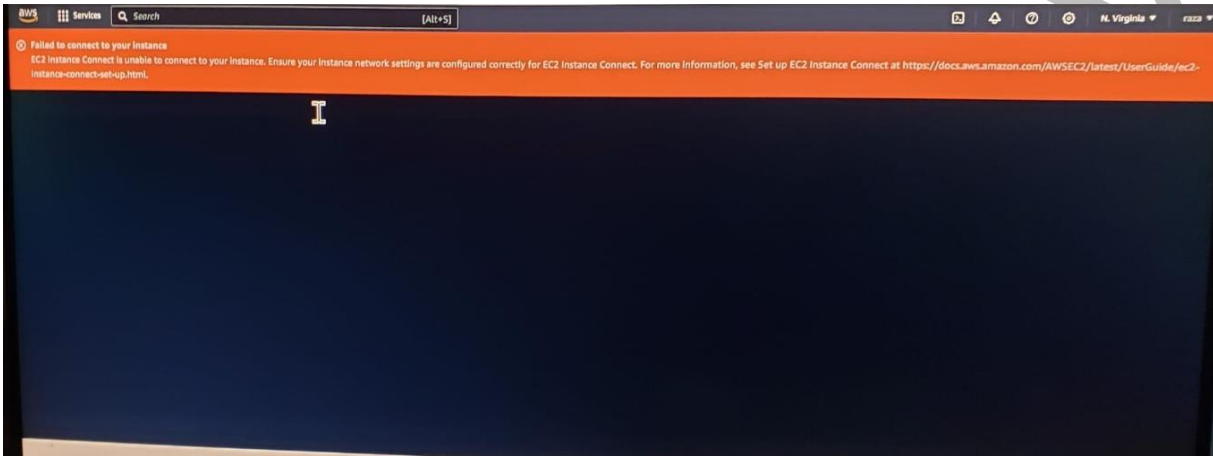
Check your key, if you are using the correct key file or not and initiate the SSH session again.

8.2 Direct Connect Option is Not working

Sometimes you can get this error message while connecting using Direct connect option

8.2.1 ERROR

Connect is unable to connect to your instance. Ensure your instance network settings are configured correctly for EC2 Instance Connect.



8.2.2 CAUSE

This is due to Networking issue. The default VPC in which EC2 instance was created does not have the internet connectivity.

8.2.3 FIX

Check the default Internet Gateway or Create the Internet Gateway and attach it to your VPC.

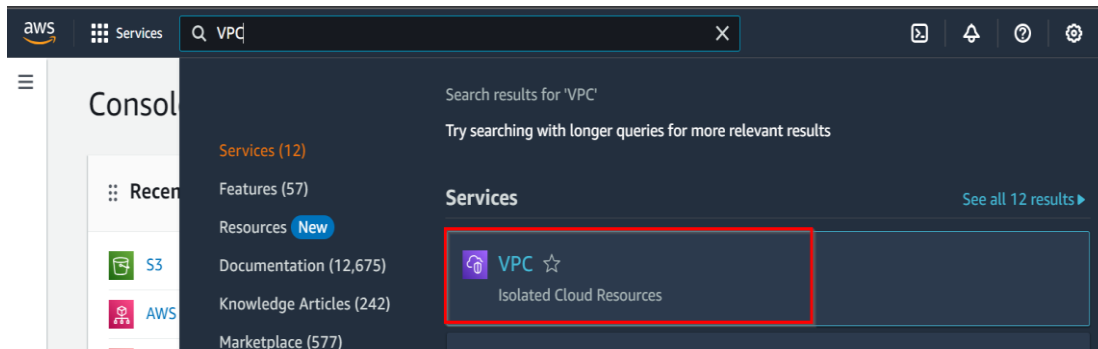
Or

To fix this issue please delete the existing Default VPC and create the New Default VPC, follow the bellow mentioned to create the Default VPC:

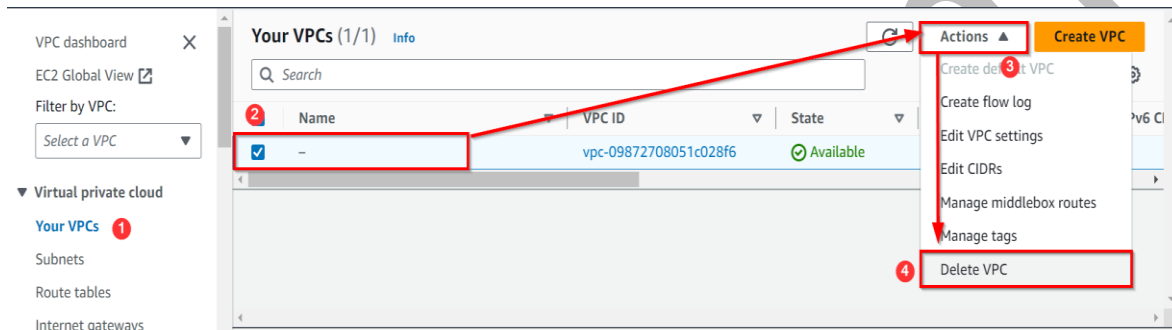
To create a default VPC using the console

1. Open the Amazon VPC console at <https://console.aws.amazon.com/vpc/>

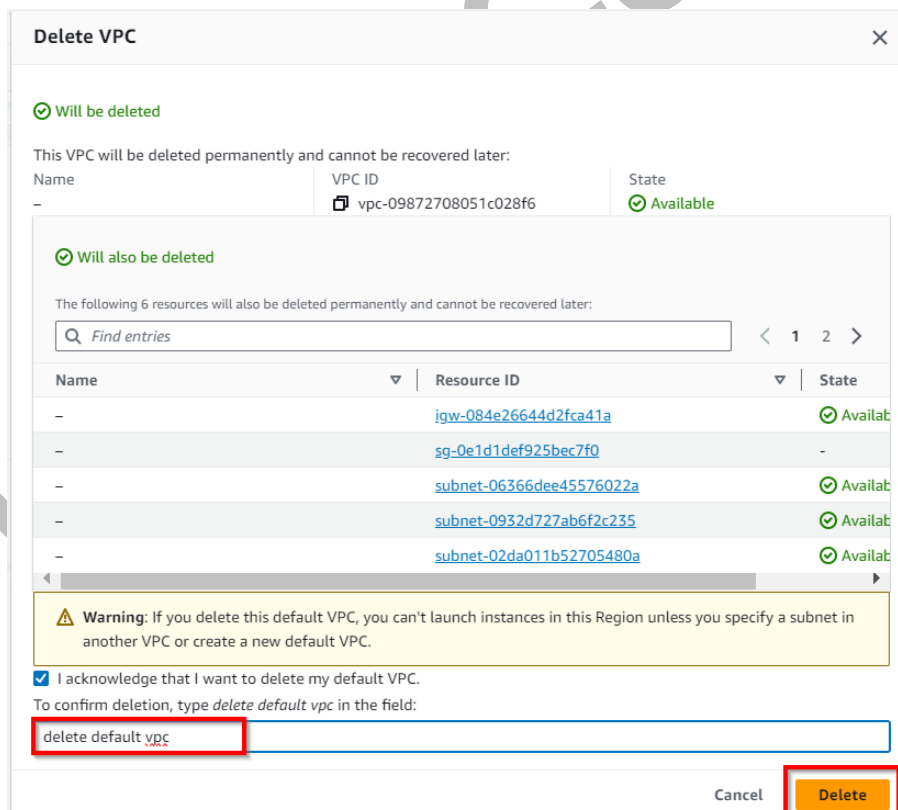
2. Search VPC in search bar and select it.



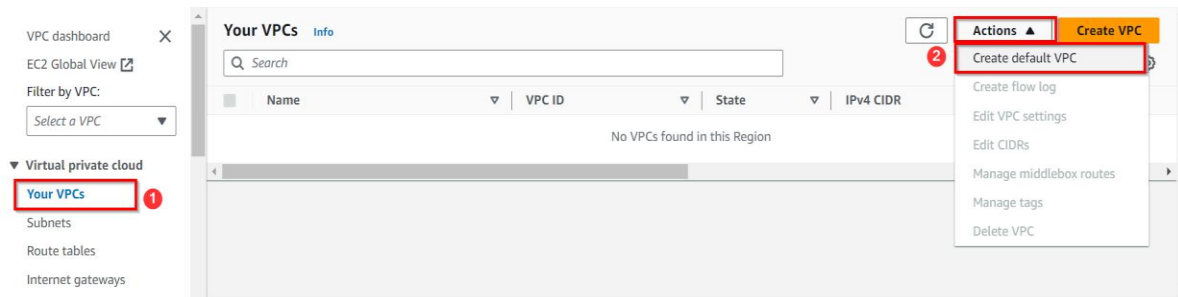
3. Delete the existing default VPC.



4. Now confirm the deletion

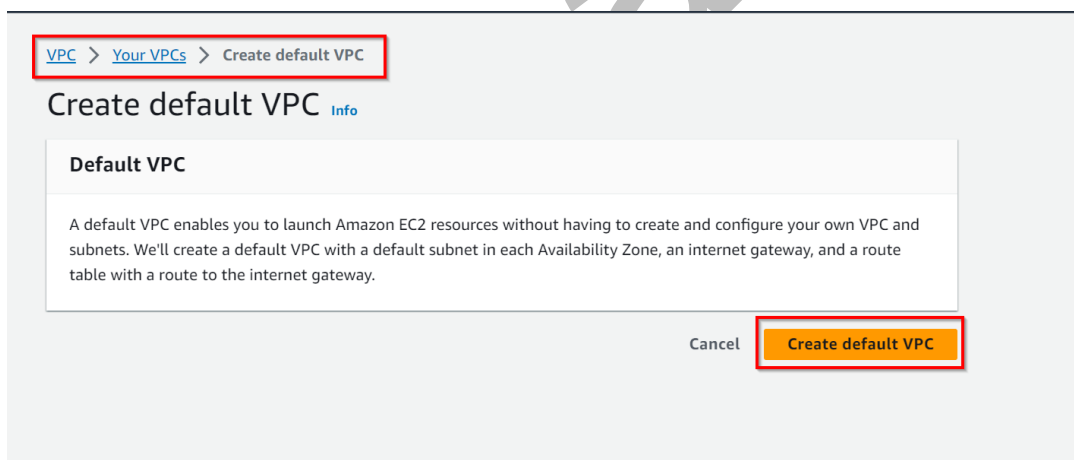


5. Go to Action.
6. Under the Action, click on create Default VPC.



A default VPC enables you to launch Amazon EC2 resources without having to create and configure your own VPC and subnets. We'll create a default VPC with a default subnet in each Availability Zone, **an internet gateway**, and a **route table** with a route to **the internet gateway**.

7. Now Click on create default VPC.



Now you will able to Create & connect to the EC2.

Suggestion: Try to create & connect the EC2 in another region as well.

9 SUMMARY

This activity guide covered steps to:

1. Launching a Linux Instance Using EC2
 - a. Creating an EC2 Instance
 - b. Accessing EC2 Instance Using Browser SSH Connection
 - c. Accessing EC2 Instance Using Putty
2. Deleting/Cleanup
 - a. Stopping the Linux Instance
 - b. Terminating the Linux Instance